

An AR-Based Multi-User Learning Environment for Anatomy Seminars

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ABSTRACT

Understanding the intricate and rapid changes in shape during embryonic formation is vital for medical students. Using the example of embryonic human heart development, we introduce an AR-based multi-user approach to enhance understanding and foster a participatory learning environment. Through a user-centered approach, we created a prototype accommodating two player roles and enabling multi-modal inputs to encourage dynamic group discussions. We invited four anatomy experts to evaluate three system configurations in an interdisciplinary workshop to assess the feasibility of integration into anatomy seminars. The gathered data and feedback indicate the potential of our collaborative concept for integration into the medical curriculum.

1 INTRODUCTION

Utilizing augmented reality (AR) technologies in medical education and training has demonstrated improvements in understanding and heightened learning motivation [2]. The use of these technologies contributes to the understanding of spatial relationships and leads to a new and improved learning paradigm [4]. The development of the embryonic heart presents a significant challenge in visualization, encompassing simultaneous changes in shape and position within a brief time frame. Research in this area highlights the benefits of advanced visualizations and interactions in anatomy education, which complement traditional methods and have a positive impact on student learning and thus on improving learning outcomes [6, 3]. In our previous work, we introduced the virtual reality (VR) application CardioGenesis4D [5], which established an immersive learning environment for medical students. This platform allows for the independent exploration of embryonic heart development through the manipulation of physiological motion sequences using hand interactions. Building upon this foundation, the current work aims to enhance learning opportunities by shifting our focus towards fostering collaboration and establishing a foundation for knowledge sharing in a more dynamic educational setting using AR technology. This progression represents a natural extension by integrating the social aspects of learning and bridging the gap between diverse didactic formats. We envisioned three configurations for the system and are interested in determining the optimal technical configuration concerning positioning, visualization, and input modalities. Applying a user-centered design approach, we conducted a participatory workshop with anatomy education experts of varying experience levels to evaluate the integration suitability of our approach in seminar contexts, gaining early insights and improvements for the integration in the medical curriculum during this developmental stage.

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2 SYSTEM DESIGN

Utilizing the Tilt Five (Tilt Five, Inc., USA) game board system, which employs an HMD to project AR overlays onto a retroreflective panel, we implemented our application for four headsets. Each user has controller input, enabling rotation, scaling, and markings using a ray-casting beam on the model. One headset was modified with an ergonomic head strap and Ultraleap's Stereo IR 170 Evaluation Kit (Ultraleap Ltd., USA) for touchless interaction. The software was built in Unity (Unity Software Inc., USA) on Microsoft Windows (Microsoft Corp., USA) with the latest Mixed Reality Toolkit 3 (MRTK) for utilizing UI elements combined with Ultraleap's Leap Motion Core Unity Plugin packages to realize optimal hand tracking. As mentioned above, the conceptual and technical foundation of this work is based on our previously published VR application [5]. We have implemented significant enhancements to the system, such as the addition of further developmental stages and the introduction of visualizations for inner structures. The 3D models now feature more detailed elements and dynamic labels have been strategically integrated to enhance clarity and user understanding. The AR approach presented in this paper includes two player roles with restricted or extended rights and control options. Envisioned is one moderator capable of guiding the seminar session step by step, along with up to three student roles to encourage dynamic group discussions. Each user has controller input to manipulate the 3D content, enabling rotation, scaling, and markings using a ray-casting beam on the model. The moderator has an extended set of functions, including toggling annotations, enforcing model alignment view, and locking the beam. Furthermore, they can deform the model through hand interaction and manipulate animations. Subsequently, three configurations of the game board were developed. In the first configuration, *Flat*, the game board lies on a table. In the second variant, *Poster*, the game board is aligned vertically. The third variant, *Curved*, enhances visualization by incorporating an additional retro-reflective foil wall for increased depth perception. See Fig.1 for reference.

3 USER STUDY

As the application modalities switched from single-user to multi-user, aiming to serve as both a visualization aid and teaching tool, collaboration among experts becomes crucial. To validate foundational assumptions and assess the utility of functionalities, we sought qualitative feedback and data collection during this development stage. Therefore, we conducted a workshop with experts in anatomy education to evaluate the suitability of our AR concept for a seminar setting. The study aimed to obtain early feedback in an iterative development process regarding the proposed input modalities, role allocation, and game board/player positioning. Moderated by a software developer, designer, and cardiologist, the evaluation aimed to qualitatively explore the visual and interactive qualities of the developed AR approach. Four anatomy experts (3 female and 1 male) from the Institute of Anatomy at the University of Magdeburg participated in the study. The participants, ranging from Research Assistants to Professors, reported 3 to 37 years of professional experience (Median = 14.5). They claimed teaching experience (Median = 17.3) in

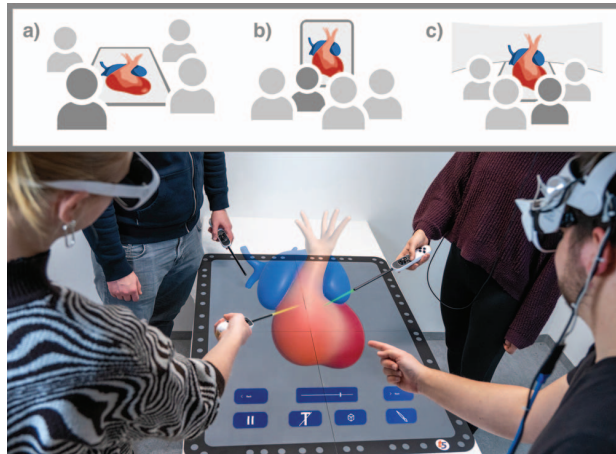


Figure 1: Top: Evaluated configurations a) *Flat*, b) *Poster*, c) *Curved*. Bottom: Overview of the multi-user AR setup in *Flat* configuration.

theoretical and practical formats such as lectures and seminars and practical training in human medicine, macro- and neuroanatomy, and histology. We employed the Technology Acceptance Model (TAM), which provides insights into user behavior and attitudes towards our approach. Through a questionnaire on a 7-point Likert scale (1 = extremely disagree; 4 = neither; 7 = extremely agree), we measured Perceived Usefulness (PU) and Perceived Ease-of-Use (PEU). These factors help gauge the effectiveness of our AR system in an educational context. Furthermore, we evaluated the visual quality of elements and user comfort using the Augmented Reality Immersion Questionnaire (ARI). This instrument measures immersion in AR settings through a 7-point Likert scale (1 = completely disagree; 4 = neutral; 7 = completely agree). We calculated an overall immersion score by averaging responses across all 21 items. To gather comprehensive feedback and improvement suggestions, experts freely expressed their thoughts and discussed them within the group. Using audio recording in a Think-aloud protocol (TAP), statements were clustered and documented if at least two matched. As for the procedure, after the welcome and consent for data recording, the technology was initially introduced. Subsequently, each concept was presented and discussed in numerical order (1st Flat, 2nd Poster, 3rd Curved). Participants went through the implemented phases of embryonic heart development, and each expert switched between the roles of moderator and student, testing different positions, input modalities, and all elements of the user interface. There was no time limit at each station. The workshop lasted around two hours and participants received no material compensation.

4 RESULTS AND DISCUSSION

The subscales of the TAM achieved a median rating of 6.1 (SD = 0.2) for PU and 5.2 (SD = 0.2) for PEU. These values indicate the general acceptance of this technology among the user group and also reflect the qualitative feedback. The ARI achieved an overall score of 5.1 (SD = 0.4), positioning it in the upper mid-range. This indicates a generally positive user experience, which correlates with a good learning environment. Ranked by experts, *Flat* took first place, followed by *Poster* in second, and *Curved* in last. The TAP also provided a series of feedback, emphasizing the effectiveness of the AR concept in leading small groups through step-by-step explanations, promoting vivid understanding, and facilitating dynamic communication. The designated role distribution and the emphasis on visualization and interaction by the moderator were

positively noted by the experts, facilitating the learning process. The test group appreciated having two input options, particularly precise pointing and controller-based rotation/scaling. Hand tracking was deemed essential for all users, enhancing motor movement and understanding physiological development. This is in line with research to assess individual learning activities [1]. The flat setting received high praise for its equal visibility and space efficiency in the seminar room. The Poster setting, while accommodating larger groups, poses potential obstructions and discomfort as users must look up. The Curved setting, providing depth but being too large for seminar integration and too distant for effective hand interaction, exhibited similar pros and cons. The standard headset's ergonomics were seen as a drawback, requiring precise positioning and causing discomfort for glasses wearers. The modified headset alleviated these issues. Visual cutoff occurred when the model extended beyond the game board. Difficulties with button operation were reported during hand interaction, especially in Poster and Curved setups. Enabling distance interaction using the controller input could enhance user experience. Partial hand-tracking issues were observed, possibly linked to headset alignment with retro-reflective material. While better placement near the game board could address this, it may limit interaction range. Seating users instead of standing could improve ergonomics and visualization. Criticisms included the four-player limit, but it was acknowledged that the learning approach is better suited for smaller groups.

5 CONCLUSION AND FUTURE RESEARCH

This work introduces an AR-based multi-user approach designed to cultivate a participatory learning environment, using embryonic heart development as an example. Through expert evaluation in three settings, we explored interaction modalities and player roles for potential integration into an anatomy seminar session. Experts suggested extending the concept to cover the development of other complex organs, like the intestine or brain. The current prototype is limited to four users simultaneously, prompting consideration for a server-based approach to increase capacity in the long term. We intend to expand to a multi-platform approach, integrating VR, and enhancing UI for optimal interaction. An evaluation involving medical students in a seminar setting is being considered. This should serve as the foundation for an in-depth analysis of the system requirements and the design of the didactic concept.

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